BAYLOR COLLEGE OF MEDICINE
CURRICULUM VITAE OUTLINE

PERSONAL STATEMENT

Each day, in hospitals across the country, patients experience life-threatening events. Their survival often depends on timely decisions of the experienced care providers who watch constantly over them. But all too often, care staff can have very little warning of impending deterioration. By the time overt symptoms appear, the damage has already been done. My research focuses on developing computer algorithms which can be used to constantly watch over patients and provide early warnings of life-threatening events. This is possible because acute decomposition does not manifest instantaneously. It builds up over a period of time, changing the physiologic dynamics of the patient in the process. These precursors can be subtle and hard for a human to see, given the level of noise in patient data, but a computer can be trained to detect such patterns. By using technology to anticipate such events in real-time, care can be provided *proactively* instead of *reactively*, which has tremendous potential for improving survival in critically ill patients. This represents a fundamental shift in the way that traditional patient monitoring and surveillance has been conducted in hospitals over the last 50 years.

My work has resulted in Sickbay, an analytics system which can process all of the data generated by all patients in a hospital system in real-time. Sickbay has been commercialized by a company that I co-founded called Medical Informatics Corp. The Sickbay system has been approved by the FDA as a 510(k) medical device and is actively being used at some of the top hospitals in the USA including Houston Methodist and Johns Hopkins. Using Sickbay at Texas Children's Hospital, my group has built several algorithms which predict critical events in the near future, the most developed of which predicts cardiorespiratory arrest in children with single ventricle physiology 1-2 hours in advance. While this is the first algorithm of its kind, it also represents a template for how clinically useful algorithms can be developed and deployed at scale. Because these algorithms require no medical grade hardware, they can be developed at a fraction of the time and cost of traditional monitoring equipment. It is my vision to see hundreds of algorithms be created to watch over millions of patients, in order to improve patient safety and outcomes in critical care environments.

I. GENERAL BIOGRAPHICAL INFORMATION

A. Personal

Craig G. Rusin
US Citizen

B. Education

Princeton University, Princeton, NJ. – *Undergraduate Student* (1997-2001)
Degree: Bachelor of Science and Engineering in the field of Chemical Engineering.
Graduation Date: June 2001

University of Virginia, Charlottesville, VA. – *Graduate Student* (2004-2009)
Degree: PhD in the field of Chemical Engineering.
Field of research: Rhythmic Engineering / Experimental Nonlinear Dynamics
Title of Dissertation: Engineering the Collective Behavior of Complex Rhythmic Systems

University of Virginia Medical School, Charlottesville, VA **-** *Post-doctoral Fellow* (2009-2010)
Division: Cardiovascular Medicine. Awarded a competitive post-doctoral fellowship by the UVA Cardiovascular Research Center which supports training for engineers for research careers in medicine.

C. Academic Appointments

Baylor College of Medicine, Houston, TX – *Associate Professor* (2011-present)
Director of the Learning Health System
Department of Pediatrics, Section of Cardiology
Department of Medicine
Head of Predictive Analytics Lab
Hospital Affiliation: Texas Children's Hospital
Field of Research**:** Medical Informatics **–** Real-time prediction of the likelihood of patient illness calculated from continuously recorded physiological data.
H-Index: 19, Total Citations: 1,200

Rice University, Houston, TX – *Adjunct Associate Professor* (2013-present)
Department: Computational Applied Mathematics and Operations Research

University of Virginia Medical School, Charlottesville, VA – *Assistant Research Professor* (2010 – 2011). Division: Cardiovascular Medicine.

D. Other Advanced Training/Experience

None

E. Other Information

E.1 Non-Academic Positions:

Arthur D. Little, Cambridge, MA - *Research Analyst* (2001 – 2004)
Worked within the Technology and Innovation Division for several Fortune 500 clients.
Arthur D. Little's Technology and Innovation Division was sold to TIAX LLC in 2002

E.1 Non-Academic Positions (cont.):

Medical Informatics Corp, Houston, TX – Co-*Founder / CTO (2011-2020), Chief Product & Innovation Officer* (2022 – Present). Medical Informatics provides a real-time remote patient monitoring and clinical analytics software platform, Sickbay™, which is utilized in clinical environments to increase quality of health care by detecting dynamic precursors of illness in real-time. Sickbay has 2M hits per month at some of the best and largest hospital systems in the USA. Sickbay covers approximately 2% of all monitored beds in the USA. Scaled company from 0 to 100 people. Raised $30M in venture capital. Obtained FDA 510K clearance for products. Obtained ISO 9001 certification for software development quality processes. Frost & Sullivan Product leader in Virtual Critical Care. Top 400 digital health companies in the world (Newsweek - 2024). Top 40 digital health analytics companies in the USA (Beckers Hospital Review - 2024).

E.2 Academic Awards

Highest Impact First Author Paper Award – Department of Pediatrics 2020
STAR Research Achievement Award from the Society of Critical Care Medicine (2018)
2015 TEDMED Hive Innovator
2012 TEDx Houston Invited Speaker
2010 SEAS Entrepreneurial Contest Winner, University of Virginia
2009 recipient of the Cardiovascular Research Center Post-doctoral Fellowship.
2009 recipient of the Chemical Engineering Faculty Award for Excellence in Doctoral Study.
2008 DuPont AIChE Catalysis and Reaction Engineering Travel Award.
Princeton University Outstanding Academic Achievement. (Fall 2000)
Procter and Gamble Design Award recipient. (Princeton University - 2001)

II. RESEARCH INFORMATION

A. Research Support

1) National Institute of Health: Pediatric Heart Network
   Award: Mentored Research Award
   Role: PI
   Time of award: 2013-2014
   Amount: $75,000

2) Seattle Children's Research Institute:
   Award: Research Subcontract
   Role: Co-Investigator
   Time of Award: 2013 - 2016

3) Foundation for Anesthesia Education and Research:
   Award: Research Fellowship Grant for Eric Vu.
   PI: Dr. Eric Vu, M.D.
   Role: Mentor
   Time of Award: 2015 - 2016

4) American Heart Association Beginning-Grant-In-Aid:
Award number: 16BGIA27490024
Title: REAL-TIME PREDICTION OF ACUTE ARREST IN INFANTS WITH SINGLE VENTRICLE PHYSIOLOGY
Role: PI
Amount: $140,000
Time of Award: 2016 – 2017

5) National Institute of Health - National Heart Lung and Blood Institute
Award number: 1R56HL131574
Title: REAL-TIME PREDICTION OF ACUTE ARREST IN INFANTS WITH SINGLE VENTRICLE PHYSIOLOGY
Role: PI
Amount: $277,123
Time of Award: 2016 - 2017

6) National Institute of Health - National Heart Lung and Blood Institute
Award number: 1R01HL142994
Title: MULTI_SITE VALIDATION STUDY OF THE HLHS ARREST PREDICTOR
Role: PI
Amount: $2,978,000
Time of Award: 2018 – 2023

6) National Institute of Health - National Institute for Child Health & Development
Award number: 1R61HD105593
Title: AICORE-KIDS: ARTIFICIAL INTELLIGENCE COVID-19 RISK ASSESSMENT FOR KIDS
Role: PI
Amount: $3,270,000
Time of Award: 2021 – 2025

7) National Science Foundation – Electrical, Communications and Cyber Systems
Award number: 2133106
Title: ASCENT: Multimodal chest e-tattoo with customized IC and deep learning algorithm for tracking and predicting progressive pneumonia
Role: co-PI
Amount: $1,500,000
Time of Award: 2021 - 2025

8) National Institute of Health - National Heart Lung and Blood Institute
Award number: 1R15LM013938
Title: Summarizing Cardiac Data: An Automated Approach for Identifying Representative Heartbeats in the Clinical Setting
Role: co-I
Amount: $375,373
Time of Award: 2022 – 2025

9) Department of Defense
Award number: W81XWH-22-C-0058
Title: Prediction and Prevention of Intracranial Hypertension and Tissue Hypoxia in Severe Traumatic Brain Injury

Role: co-I
Amount: $744,044
Time of Award: 2022 – 2025

B. National Scientific Participation

Academic Conference Organization

- Rice Data Science Conference - 2017 to present
  Member of the administrative and review committee for the Rice University Data Science Conference. This 2 day conference is held annually at Rice University and attracts speakers from both industry and academia. A diverse set of fields is represented from healthcare and oil and gas, to deep mathematical data science theory. The conference regularly draws several hundred attendees. All presentations are recorded with the speaker's permission and made available online as enduring educational materials.

Grant and Journal Reviewer Positions

- NIH Study Section - NATIONAL INSTITUTE OF DIABETES AND DIGESTIVE AND KIDNEY DISEASES SPECIAL EMPHASIS PANEL (ZDK1 GRB - G (J1))
- Netherlands Organization for Scientific Research (NWO) - Applied and Engineering Sciences (grant review)
- NIH ad-hoc reviewer (grant review)
- American Medical Informatics Association (AMIA) for National Conference.
- Journal of Biomechanical Engineering (journal review)
- PLOS One (journal review)
- PLOS Computational Biology (journal review)
- Pediatrics (journal review)
- Computers in Biology and Medicine (journal review)

Invited Talks

1. C.G. Rusin, et. al. " Operationalizing Predictive Analytics in Critical Care Environments", Health Analytics Summit (HAS), Salt Lake City, UT (Sept 2018)

2. C.G. Rusin, et. al. " Predictive Analytics and Software as a Medical Device", Consumer Electronics Show (CES), Las Vegas, NV (Jan 2018)

3. C.G. Rusin "Predictive Analytics in Critical Care Environments", Rice ML. Rice University, Houston, TX (Feb 2017).

4. C.G. Rusin "Predictive Analytics in the CVICU" Children's Mercy Cardiac Forum. Kansas City, MO (Oct. 2016).

5. C.G. Rusin "Real-Time Clinical Decision Support Technologies" American Society of Health-System Pharmacists (ASHP) Semi-Annual Meeting. Denver, CO (June 2015).

6. C.G. Rusin "Physiological Monitoring" Guest lecturer, Nationwide Children's Hospital. (April 2014).

7. C.G. Rusin "Physiological Monitoring" Guest lecturer, Seattle Children's Research Institute. (Sept 2013).

8. C.G. Rusin "Physiological Monitoring Guest lecturer, MIT, Harvard-MIT Division of Health Sciences and Technology. (Nov 2010).

9. C.G. Rusin "Engineering the collective behavior of rhythmic Chemical and Biological Systems" Guest lecturer, Technical University of Berlin, Department of Biological Engineering. (April 2009).

10. C.G. Rusin "Engineering the collective behavior of rhythmic Chemical and Biological Systems" Invited colloquium speaker, College of William and Mary, Department of Physics. (Oct. 2009).

Conference Presentations

1. Elizondo LI, Rios DR, Vu E, Kibler KK, Easley RB, Andropoulos DB, Acosta S, Rusin CG, Kaiser JR, Brady K, Rhee CJ. "Observed and calculated cerebral critical closing pressure are highly correlated in preterm infants." Pediatric Academic Societies, May 2018.

2. Vu E, Rusin CG, Dreyer W, Devaraj S, Easley RB, Andropoulos D, Brady K . "ST segment variability in coronary allograft vasculopathy after pediatric heart transplantation." Association of University Anesthesiologists and International Anesthesia Research Society Moderated Poster Presentation. Chicago, IL, April 2018.

3. Elizondo LI, Rios DR, Vu E, Kibler KK, Easley RB, Andropoulos DB, Acosta S, Rusin CG, Kaiser JR, Brady K, Rhee CJ. Observed and calculated cerebral critical closing pressure are highly correlated in preterm infants. NeoHeart Conference, March 2018.

4. Vu E, Rusin CG, Dreyer W, Devaraj S, Easley RB, Andropoulos D, Brady K.. "A pilot study of biomarkers for the detection of coronary allograft vasculopathy after pediatric heart transplant ". Society for Pediatric Anesthesia Conference and Congenital Cardiac Anesthesia Society Oral Presentation. Phoenix, AZ. March 2018.

5. Achuff BJ, Moffett, B, Acosta, S; Lasa, J; Checchia, PA; Rusin, CG: Hypotensive Response to IV Acetaminophen in Pediatric Cardiac Patients; Poster presentation Pediatric Critical Care Society, Washington DC, Dec 2017.

6. Vu E, Rusin CG, Easley RB, Kibler K, Andropoulos D, Brady K.."Blood pressure management in hypoplastic left heart syndrome after stage 1 palliation: Determination of an ST instability-dependent optimal blood pressure." Association of University Anesthesiologists and International Anesthesia Research Society Moderated Poster Presentation. Washington, DC. May 2017.

7. Rhee CJ, Rios DR, Vu E, Kibler KK, Easley RB, Andropoulos DB, Rusin CG, Kaiser JR, Brady KM. "Periventricular leukomalacia is associated with deviation below optimal arterial blood pressure." Pediatric Academic Societies, May 2017.

8. Cruz SM , Lau PE, Cass DL, Style CC, Rusin CG, Lee TC, Rhee CJ, Keswani S, Ruano R, Welty SE, Olutoye OO." A Novel Multimodal Computational System Using Near-infrared Spectroscopy Predicts the Need for ECMO Initiation in Neonates with Congenital Diaphragmatic Hernia". American Pediatric Surgical Association Abstract. 2017.

9. Cruz SM, Lau PE, Cass DL, Style CC, Rusin CG, Lee TC, Rhee CJ, Keswani S, Ruano R, Welty SE, Olutoye OO. "A novel multimodal computational system using near-infrared spectroscopy predicts the need for ECMO initiation in neonates with congenital diaphragmatic hernia." Texas Children's Hospital Pediatric Research Day, March 2017.

10. Smith B., Vu E., Kibler KK, Rusin CG., Easley RB, Andropoulos D, Heinle J., Czosnyka M, Licht D., Lynch J, Brady KM. "Does profound hypothermia during cardiopulmonary bypass impair autoregulation in neonates?" American Society of Anesthesia National Meeting, Chicago, IL October 22 - 26, 2016.

11. Smith B, Vu E, Kibler KK, Rusin CG, Easley RB, Andropoulos D, Heinle J, Czosnyka M, Licht D, Lynch J, Brady KM "Where is the optimal ABP for an infant supported by cardiopulmonary bypass? American Society of Anesthesia National Meeting, Chicago, IL October 22 - 26, 2016.

12. Achuff BJ, Rusin CG, et al. "Epinephrine Syringe Exchange Events in a Pediatric Cardiovascular Critical Care Unit: Analyzing the Storm" 8th World Congress on Pediatric Intensive and Critical Care, Toronto, Canada, June 4-8, 2016.

13. Vu E, Rusin CG, Penny DJ, Kibler K, Easley RB, Smith B, Andropoulos D, Brady K. "ST segment variability is an improved predictor of clinical deterioration in single ventricle patients compared to standard ST segment monitoring." Association of University Anesthesiologists Moderated Poster Presentation. San Francisco, CA. May 2016.

14. Rusin CG, Brady K, Penny D. "Data Collection in the CVICU: When is Enough, Enough?" Oral Presentation, Pediatric Cardiovascular Intensive Care Society, Houston, TX. December 2015.

15. Rhee CJ, Kibler KK, Easley RB, Andropoulos DB, Rusin CG, Brady KM, Kaiser JR. "The impact of hypercapnea on systolic cerebral autoregulation and the development of intraventricular hemorrhage in premature infants." Pediatric Academic Societies, May 2015.

16. Vu E, Rusin CG, Penny DJ, Kibler K, Easley RB, Smith B, Andropoulos D, Brady K. "ST segment variability as a predictor of clinical deterioration in single ventricle patients." Congenital Cardiac Anesthesia Society Moderated Poster Presentation. Phoenix, AZ. March 2015.

17. Vu E, Rusin CG, Penny DJ, Kibler K, Easley RB, Smith B, Andropoulos D, Brady K. "ST segment variability as a predictor of clinical deterioration in hypoplastic left heart syndrome." American Heart Association Moderated Poster Presentation. Chicago, IL. November 2014.

18. Rusin CG, Vu E. Brady K. Penny DJ. "Prediction of Imminent Deterioration of Children With Parallel Circulations Using Real-Time Processing of Physiological Data" Oral Presentation, American Heart Association Scientific Session, Chicago IL. (Nov 2014).

19. Rhee CJ, Fraser 3rd CD, Kibler KK, Easley RB, Andropoulos DB, Czosnyka M, Varsos GV, Smielewski P, Rusin CG, Brady KM, Kaiser JR. "The ontogeny of cerebrovascular critical closing pressure in human premature infants." Pediatric Academic Societies, May 2014.

20. Rhee CJ, Fraser 3rd CD, Kibler KK, Easley RB, Andropoulos DB, Czosnyka M, Varsos GV, Smielewski P, Rusin CG, Brady KM, Kaiser JR. "The ontogeny of cerebrovascular pressure autoregulation in human premature infants." Pediatric Academic Societies, May 2014.

21. Rhee CJ, Fraser 3rd CD, Kibler KK, Easley RB, Andropoulos DB, Czosnyka M, Varsos GV, Smielewski P, Rusin CG, Brady KM, Kaiser JR. "Diastolic collapsing margin is associated with intraventricular hemorrhage in premature infants." Pediatric Academic Societies, May 2014.

22. Rusin CG. "Engineering the Collective Behavior of Soft Matter System Using Nonlinear Time-delayed Feedback Stimulation. Application to Biological Neurons" Dynamics Days Conference, Oldenburg, Germany. September 2011.

23. Lee H., Delos J, Pagent-Brown A, Rusin CG, Lake D, Moorman JR, "A New Algorithm for Detection of Apnea in Infants in Neonatal Intensive Care Units." Pediatric Academic Society National Conference, Vancouver BC, Canada May 2011.

24. Hudson JL, Rusin CG, et. al. "Complexity in Rhythmic Chemical Systems." Workshop on oscillation dynamics in neural & biochemical systems. University of Exeter. September 2010.

25. Hudson JL, Rusin CG. "Collective Dynamic Behavior of Coupled Chemical Oscillators." National Conference of the American Institute of Chemical Engineering. Salt Lake City, UT. November 2010.

26. Vergales B, Pagent-Brown A, Callahan S, Rusin CG, Lake D, Moorman J., Kattwinkel J. "Apnea of Prematurity – A Comparison of Nursing Records with Continuous Electronic Waveform Analysis." Pediatric Academic Society National Conference, Vancouver BC, Canada. May 2010.

27. Rusin CG, Kattwinkel J, Paget-Brown A, Lake D, Moorman J, "Neonatal Apnea Alarms Increase During Early Morning Hours" Pediatric Academic Society National Conference, Vancouver BC, Canada. May 2010.

28. Rusin CG, Clark M, Kattwinkel J, Lake D, Moorman J. "Cluster Computing Data Center for Recording and Parallel Processing of Cardiovascular Waveforms", 6th conference of the European Study Group on Cardiovascular oscillations. Humbolt University, Berlin, Germany. April 2010.

29. Rusin CG, Kiss I, Kori H, Hudson J. "Engineering Complexity in Chemically-Reacting Dynamical Systems" National Conference of the American Institute of Chemical Engineering. Philadelphia, PA. Nov. 2008.

30. Rusin CG, Kiss I, Kori H, Hudson J. "Engineering Collective Behaviors in Reacting Dynamical Systems" Graduate Research Seminar Series. University of Virginia, Charlottesville, VA. June 2008.

31. Hudson J, Kiss I, Rusin CG, Zhai Y. "Complexity in Rhythmic Electrochemical Systems." 211th Conference of the Electrochemical Society, Symposium: "Multi-Scale Simulations of Electrochemical Systems." Chicago, IL. May 2007.

32. Kiss I, Rusin CG, Hudson J. "Dynamical Order and Complexity in Rhythmic Chemical Systems." 3rd International IEEE Scientific Conference on Physics and Control. Potsdam, Germany. Sept. 2007.

33. Kiss I, Rusin CG. Hudson, J. "Engineering Complex Dynamical Structures in Electrochemical Systems: Clustering, Sequential Patterns, and Desynchronization." Gordon Research Conference. Colby College, Waterville, ME. July 2008.

34. Hudson J., Rusin CG, Kiss I. "Engineering Complex Dynamical Structures in Populations of Chemical Oscillators." 4th International Symposium: Engineering of Chemical Complexity. Fritz-Haber Institute, Berlin, Germany. May 2008.

C. Publications

1. I.Z. Kiss, C.G. Rusin, H. Kori, and J.L. Hudson. "Engineering Complex Dynamical Structures: Sequential Patterns and Desynchronization." *Science* **316,** 5833, 1886-1889 (**2007**). Work done at UVA.

2. H. Kori, C.G. Rusin, I.Z. Kiss, and J.L. Hudson, "Synchronization Engineering: Theoretical Framework and Application to Dynamical Clustering." *Chaos* **18**, 2, 026111 (**2008**). Work done at UVA.

3. C.G. Rusin, I.Z. Kiss, H. Kori, and J.L. Hudson, "Framework for Engineering the Collective Behavior of Complex Rhythmic Systems." *Ind. Chem. Eng. Res.* **48**, 21, 9416-9422 (**2009**). Work done at UVA.

4. C.G. Rusin, H. Kori, I.Z. Kiss, and J.L. Hudson, "Synchronization Engineering: Tuning the Phase Relationship Between Dissimilar Oscillators Using Nonlinear Feedback." *Phys. Trans. R. Soc. A* 368, 1918, 2189-2204 (**2010**). Work done at UVA.

5. C.G. Rusin, I. Tokuda, I.Z. Kiss, and J.L. Hudson. "Engineering of Synchronization and Clustering of a Population of Chaotic Chemical Oscillators." *Angewandte Chemie*, DOI: 10.1002/ange.201008194. (**2011**). Work done at UVA.

6. K. A. Blaha, A. Pikovsky, M. Rosenblum, M.T. Clark, C.G. Rusin, and J. L. Hudson. "Reconstruction of 2-D phase dynamics from experiments on coupled oscillators." *Phys. Rev. E*. 84, 4 (**2011**) DOI: 10.1103/PhysRevE.84.046201. Work done at UVA.

7. C.G. Rusin, S.E. Johnson, J. Kapur, and J.L Hudson. "Controlling the Synchronization of Neuron Action Potentials Using Global Time-delayed Feedback Stimulation." *Phys. Rev. E*, **84**, 066202 (**2011**). Work done at UVA.

8. M.T. Clark, C.G. Rusin, J.L. Hudson, H. Lee, J.B. Delos, L. Guin, T. Smoot, B. Vergales, A.P. Brown, J. Kattwinkel, D. Lake, J.R. Moorman, "Cardiorespiratory Coupling in Premature Infants: Breath-by-Breath Analysis of Respiratory Sinus Arrhythmia." *J. Appl. Physiol*. jap.01152.2011; (**2011**). Work done at UVA.

9. M.T. Clark, C.G. Rusin, J.L. Hudson, H. Lee, J.B. Delos, L. Guin, T. Smoot, B. Vergales, A.P Brown, J. Kattwinkel, D. Lake, R. Moorman. "Acute uncoupling of heartbeat and respiration in neonatal respiratory decompensation." J. Crit. Care. **27**, 3, 1. (**2012**). Work done at UVA.

10. H. Lee, C.G. Rusin, D. Lake, A.P. Brown, B.D. Vergales, J. Kattwinkel, J.R. Moorman, J.B. Delos "A new algorithm for detecting central apnea in neonates." *Physiol. Meas*. **33**, 1-17 (**2012**). Work done at UVA.

11. C. Hu, C.G. Rusin, Z.Tan, N.A. Guagliardo, and P.Q. Barrett, "Zona glomerulosa cells of the mouse adrenal cortex are intrinsic electrical oscillators." *J. Clin. Invest*. **122**, 6, 2046-53 (**2012**). Work done at UVA.

12. K. Zagol, D.E. Lake, B. Vergales, M.E. Moorman, A. Paget-Brown, H. Lee, C.G. Rusin, J.B. Delos, M.T. Clark, J.R. Moorman, J. Kattwinkel, "Anemia, Apnea of Prematurity, and Blood Transfusions." *J. Pediatrics* 161, 3, 417-421. doi:10.1016/j.jpeds.2012.02.044 (**2012**). Work done at UVA.

13. K. Brady, R. Easley, K. Kibler, D. Kaczka, D. Andropoulos, C. Fraser, P. Smielewski, M. Czosnyka, G. Adams, C. Rhee, and C. G. Rusin, "Positive End-Expiratory Pressure Oscillation Facilitates Brain Vascular Reactivity Monitoring" *J. Appl. Physiol.* **113**, 9, 1362-1368 (**2012**). Work done at BCM.

14. C.D. 3rd Fraser, K.M. Brady, C.J. Rhee, R.B. Easley, K. Kibler, P. Smielewski, M. Czosnyka, D.W. Kaczka, D.B. Andropoulos, C.G. Rusin. "The frequency response of cerebral autoregulation*." J Appl Physiol*. 115, 1, 52. (**2013**). Work done at BCM.

15. A.D. Politano, L.M. Riccio, D.E. Lake. C.G. Rusin, L.E. Guin, C.S. Josef, M.T. Clark, R.G. Sawyer, J.R. Moorman, J.F. Calland. "Predicting the Need for Urgent Intubation in a Surgical/Trauma Intensive Care Unit." *Surgery* 154, 5, 1110 (**2013**) . Work done at UVA.

16. B.D. Vergales, A.O. Paget-Brown, H. Lee, L.E. Guin, T.J. Smoot, C.G. Rusin, M.T Clark, J.B. Delos, K.D. Fairchild, D.E. Lake, R. Moorman, J. Kattwinkel. "Accurate automated apnea analysis in preterm infants." *Am. J. Perinatol*. 31, 2, 157 (**2014**). Work done at UVA.

17. C.J. Rhee, C.D. Fraser III, K. Kibler, R.B. Easley, D.B. Andropoulos, M. Czosnyka, G.V. Varsos, P. Smielewski, C.G. Rusin, K.M. Brady, J.R. Kaiser. "The Ontogeny of Cerebrovascular Pressure Autoregulation in Premature Infants*". J. Perinatology* 1-6 DOI:10.1038/jp.2014.122 (**2014**). Work done at BCM.

18. S. Acosta, C. Puelz, B. Riviere, D.J. Penny, C.G. Rusin. "Numerical Method of Characteristics for One-Dimensional Blood Flow." *J. Comp. Phys*. 294, 96 (**2015**). Work done at BCM.

19. S. Acosta, D.J. Penny, C.G Rusin. "An effective model of blood flow in capillary beds" *Microvascular Research* 100, 40-47 (**2015**). Work done at BCM.

20. C.J. Rhee, C.D. Fraser, K.K. Kibler, R.B. Easley, D.B. Andropoulos, M. Czosnyka, G.V. Varsos, P. Smielewski, C.G. Rusin, K..M. Brady, J.R. Kaiser. "Ontogeny of cerebrovascular critical closing pressure." *Peds. Research* 78, 1, 20 71-75 (**2015)**. Work done at BCM.

21. S. Cruz, A.C. Akinkuotu, C.G. Rusin, D.L Class, T.C. Lee, S.E. Welty, O.O Olutoye "A Novel Multimodal Computational System Using Near-infrared Spectroscopy to monitor Cerebral Oxygenation during Assisted Ventilation in CDH patients." *J. Peds. Surg.* 51, 1, 38-43 (**2016**). Work done at BCM.

22. C. G. Rusin, S. Acosta, E.L. Vu, A.C. Bavare, R.B. Myers, L.W. Patterson, L.S. Shekerdemian, K. M. Brady, Daniel J. Penny "Prediction of Imminent Deterioration of Children With Parallel Circulations Using Real-Time Processing of Physiological Data" *J. Thorac. Cardio. Surg*. 152(1) 171-177 (**2016**). Work done at BCM.

23. R.B. Myers, C. Lazaridis, C.M Jermaine, C.S. Robertson, C.G. Rusin, " Predicting Intracranial Pressure and Brain Tissue Oxygen Crises in Patients with Severe Traumatic Brain Injury." *Crit. Care Med*. 44, 9 1754-1761 (**2016**). Work done at BCM.

24. D.R. Busch, C.G Rusin, D.J. Licht, K. Brady, J Heinle, K.K. Kibler, W.M. Hance. "Continuous cerebral hemodynamic measurement during deep hypothermic circulatory arrest" *Biomed Opt Express* 7, 9 3461-3470 (**2016**). Work done at BCM.

25. C.J. Rhee, C.D. Fraser III, K. Kibler, R.B. Easley, D.B. Andropoulos, M. Czosnyka, G.V. Varsos, P. Smielewski, C.G. Rusin, K.M. Brady, J.R. Kaiser. "Elevated diastolic closing margin is associated with intraventricular hemorrhage in premature infants*". J. Pediatrics*. 174 52-56 (**2016**). Work done at BCM.

26. S.M. Cruz, P.E. Lau, C.G. Rusin, C.C. Style, D.L. Cass, C.J. Fernandes, T.C. Lee, C.J. Rhee, S. Keswani, R. Ruano, S.E. Welty, O.O. Olutoye. "A novel multimodal computational system using near-

infrared spectroscopy predicts the need for ECMO initiation in neonates with congenital diaphragmatic hernia." *J. Peds Surg.* **(2017)**. Work done at BCM.

27. E.L. Vu ,C.G. Rusin, D.J. Penny, K.K. Kibler, R.B. Easley, B. Smith, D. Andropoulos, K.M. Brady. "A Novel Electrocardiogram Algorithm Utilizing ST-Segment Instability for Detection of Cardiopulmonary Arrest in Single Ventricle Physiology: A Retrospective Study." *Ped. Crit.Care Med.* 18,1 44-53 (**2017**). Work done at BCM.

28. C. Puelz, S. Čanić, B. Rivière, C.G. Rusin. "Comparison of reduced models for blood flow using Runge–Kutta discontinuous Galerkin methods." *Appl. Num. Math.* 115-1 114-141 (**2017**). Work done at BCM.

29. B. Smith, E. Vu, K. Kibler, C.G. Rusin, R.B Easley, D. Andropoulos, J. Heinle, M Czosnyka, D. Licht, J. Lynch, K.M. Brady. "Does hypothermia impair cerebrovascular autoregulation in neonates during cardiopulmonary bypass?" *Pediatric Anesthesia*. 9, 905-910 **(2017).** Work done at BCM.

30. C. Puelz, S. Acosta, B. Rivière, D.J. Penny, K.M. Brady, C.G Rusin. "A computational study of the Fontan circulation with fenestration or hepatic vein exclusion CUR paper". *Computers in Biology and Medicine*. 89, 1, 405-418. **(2017)**. Work done at BCM.

31. S. Acosta, C. Puelz, B. Rivière, D.J. Penny, K.M. Brady, C.G. Rusin. "Cardiovascular mechanics in the early stages of pulmonary hypertension: a computational study." *Biomech Model Mechanobiol*. **(2017)**. Work done at BCM.

32. B.J. Achuff, J.C. Achuff, H.H Park, B. Moffett, S. Acosta, C.G. Rusin, P.A. Checchia. "Epinephrine syringe exchange events in a paediatric cardiovascular ICU: analysing the storm." *Cardiology in the Young.*28,3, 409-415 **(2018).** Work done at BCM.

33. S. Acosta, D.J. Penny, K.M. Brady, C.G. Rusin. "An effective model of cerebrovascular pressure reactivity and blood flow autoregulation.". *Microvasc. Res*. 115:34 **(2018)**. Work done at BCM.

34. E.P. Hendryx, B.M. Rivière, D.C. Sorensen, C.G. Rusin. "Finding representative electrocardiogram beat morphologies with CUR." *J. of Biomed. Informatics* 77, 97-110 **(2018).** Work done at BCM.

35. S.E. Welty, C.G. Rusin, L.I. Stanberry, G.T. Mandy, A.I. Gest, J.M. Ford, C.H. Backes, P.C. Richardson, C.R. Howard, T.N. Hansen, C.V. Smith. "Short term evaluation of respiratory effort by premature infants supported with bubble nasal continuous airway pressure using Seattle-PAP and a standard bubble device." *PLos ONE*, 13,3 **(2018).** Work done at BCM.

36. S.O. Valdes, C.Y. Miyake, M.C. Niu, C.M. Uz, S.Y. Asaki, A.P. Landstrom, A.E. Schneider, C.G. Rusin, R. Patel, W.W. Lam, J.J. Kim. "Early Experience with Intravenous Sotalol in Children with and without Congenital Heart Disease" *Heart Rhythm* 15,12, 1862-1869 (**2018**). Work done at BCM.

37. M.A. Hussein, N. Deng, C.G. Rusin, E.E. Paysse, A. Bhatt, D.K. Coats. "Heart rate variability changes and its association with the development of severe retinopathy of prematurity." J. AAPOS. 22, 5, 371-375 (**2018**). Work done at BCM.

38. S.M Cruz, P.E. Lau, C.G. Rusin, C.C. Style, D.L. Cass, C.J. Fernandes, T.C. Lee, C.J. Rhee, S. Keswani, R. Ruano, S.E. Welty, O.O. Olutoye. "A novel multimodal computational system using near-infrared spectroscopy predicts the need for ECMO initiation in neonates with congenital diaphragmatic hernia." *J. Ped. Surg*. 53,1, 152-158 (**2018**). Work done at BCM.

39. C. Lazaridis, C.G. Rusin, C.S. Robertson. "Secondary Brain Injury: Predicting and Preventing Insults." *Neuropharmacology* 145, 145-152 (**2019**). Work done at BCM.

40. B.J. Achuff, B.S. Moffett, S. Acosta, J.J. Lasa, P.A. Checchia, C.G. Rusin. "Hypotensive Response to IV Acetaminophen in Pediatric Cardiac Patients" *Peds. Crit. Care Med.* 20, 6, 527-533 (**2019**). Work done at BCM.

41. C.G. Rusin, J.J. Lasa, P.A. Checchia. "Commentary: The patient is the focus, but the data are the key: Toward data-driven critical care environments" *J. Thor. Cardio. Surg*. 158,1, 244-245 (**2019**). Work done at BCM.

42. L.I. Elizandro, D.R. Rios, E. Vu, K. Kibler, R.B. Easley, D.B. Andropoulos, S. Acosta, C.G. Rusin, J.R. Kaiser, K.M. Brady, C.J. Rhee. "Observed and calculated cerebral critical closing pressure are highly correlated in preterm infants" *Peds. Res.* 86, 2, 242-246 (**2019**). Work done at BCM.

43. S. Acosta, M. Ahmed, S.M. Yin, K.M. Brady, D.J. Penny, C.G Rusin. "A robust Fourier-based method to measure pulse pressure variability" Biomed. Sig. Proc. & Control. 60, 101947 (**2020**). Work done at BCM.

44. N.A. Guagliardo, P.M. Klein, C.A. Gancayco, A. Lu, S. Leng, R.R. Makarem, C. Cho, C.G. Rusin, D.T. Breault, P.Q. Barrett, M.P Beenhakker. "Angiotensin II induces coordinated calcium bursts in aldosterone-producing adrenal rosettes." Nature Comm. 11,1, 1679 (**2020**). Work done at BCM.

45. D.R. Rios, C.J. Rhee, L. Elizondo, K.M. Brady, C.G. Rusin, S. Acosta. "Creatinine filtration kinetics in critically Ill neonates" *Pediatr Res.* 89,4. (**2021**). Work done at BCM.

46. E.P. Hendryx, B.M. Rivière, C.G. Rusin. "An Extended DEIM Algorithm for Subset Selection and Class Identification." *Mach Learn.* 110,4,621-650. (**2021**). Work done at BCM.

47. Z.C. Foughty, O. Tavaslioglu, C.J. Rhee, L.I Elizondo, C.G. Rusin, D.J. Penny, S. Acosta, D.R. Rios. "Novel Method of Calculating Pulse Pressure Variation to Predict Fluid Responsiveness to Transfusion in Very Low Birth Weight Infants." *J Pediatr*. (**2021**). Work done at BCM.

48. C.G. Rusin, S.I. Acosta, E.L. Vu, M. Ahmed, K.M. Brady, D.J. Penny. "Automated Prediction of Cardiorespiratory Deterioration in Patients with Single Ventricle." *J Am Coll Cardiol*. 77,25,3184-3192. (**2021**). Work done at BCM.

49. J.L.S. Waugh, R. Patel, Y. Ju, A.B. Patel, C.G. Rusin, P.N. Jain. "A novel automated junctional ectopic tachycardia detection tool for children with congenital heart disease." *Heart rhythm O2.* 3,3, 302-310. (**2022**). Work done at BCM.

50. P.N. Jain, S. Acosta, A. Annapragada, P.A. Checchia, A. Moreira, E. Muscal, S.E. Sartain, S.K.S. Tejtel, T.P. Vogel, L. Shekerdemian, C.G. Rusin. "Comparison of Laboratory and Hemodynamic Time Series Data Across Original, Alpha, and Delta Variants in Patients with Multisystem Inflammatory Syndrome in Children." *Pediatr Crit Care Med*. May 04 (**2022**). Work done at BCM.

51. F. Savorgnan, S. Acosta, A. Alali, A. Moreira, A. Annapragada, C.G. Rusin, S. Flores, R.S. Loomba, A. Moreira. "Social and Demographic Disparities in the Severity of Multisystem Inflammatory Syndrome in Children." Pediatr *Infect Dis J*. 06 01; 41(6):256-258. (**2022**). Work done at BCM.

52. F. Savorgnan, A. Moreira, A. Moreira, A. Annapragada, S.K.S. Tejtel, S. Flores, C.G. Rusin, L. Shekerdemian, S. Acosta. "Physiologic profile associated with severe multisystem inflammatory syndrome in children: a retrospective study." *Pediatr Res*. May 14. (**2022**). Work done at BCM.

53. Z. Ahmad, L.H. Jin, D.J. Penny, C.G. Rusin, C.S. Peskin, C. Pulz. "Optimal Fenestration of the Fontan Circulation" *Frontiers in Phys.*13 (**2022**). Work done at BCM.

54. S.W. Han, C. Puelz, C.G. Rusin, D.J. Penny, R. Coleman, C.S. Peskin. "Computer simulation of surgical interventions for the treatment of refractory pulmonary hypertension." *Mathematical Med and Bio.* (**2022**). Work done at BCM.

55. D. Lior, C. Puelz, C. Edwards, S. Molossi, B.E. Griffith, R.K Birla, C.G. Rusin. "Semi-Automated Construction of Patient-Specific Aortic Valves from Computed Tomography Images" *Annl. Biomed Eng.* 51,1, 189-199. (**2023**). Work done at BCM.

56. F. Savorgnan, R.S. Loomba, S. Flores, C.G. Rusin, F. Zheng, A.M, Hassan, S. Acosta. "Descriptors of Failed Extubation in Norwood Patients Using Physiologic Data Streaming" *Peds. Cardio*. 44,2, 396-403. (**2023**). Work done at BCM.

57. P.S. Connell, JF Price, CG Rusin, TS Howard, JA Spinner, SO Valdes, TDN Pham, CY Miyake, JJ Kim, JJ. "Decreased Heart Rate Variability in Children with Acute Decompensated Heart Failure is Associated with Poor Outcomes." *Pediatric Cardiology*. 45, 8 1767-1775. **(2024).** Work done at BCM.

58. MB Nguyen, M Venet, CPS Fan, A Dragulescu, CG Rusin, LL Mertens, S Mital, O Villemain. "Modeling the Relationship Between Diastolic Phenotype and Outcomes in Pediatric Hypertrophic Cardiomyopathy." *J. American Soc. Echocardiography*. 37, 5. **(2024)**. Work done at BCM.

59. YL Ju, JLS Waugh, S Singh, CG Rusin, AB Patel, PN Jain. "A multimodal deep learning tool for detection of junctional ectopic tachycardia in children with congenital heart disease." *Heart Rhythm O2*. 5, 7, 452-459. **(2024).** Work done at BCM.

60. C Puelz, CG Rusin, D Lior, S Sachdeva, TT Doan, LF Eilers, D Reaves-O'Neal, M Akula, S Molossi. "Fluid-structure interaction simulations for the prediction of fractional flow reserve in pediatric patients with anomalous aortic origin of a coronary artery." *Med. Engineering & Physics*. 136 **(2025).**

D. Innovation and Commercialization

Patents

1. Rusin CG, Brady KM, Vu E, Acosta S, Penny DJ. "CLINICAL METRIC FOR PREDICTING ONSET OF CARDIORESPIRATORY DETERIORATION IN PATIENTS. Baylor College of Medicine. United States of America, assignee. (May 2016).

2. U.S. provisional patent # 60/937,595, Supplemental provisional patent # 61/054,861, "Mild Stimulation Device and Measurement Based Design Method for Non-Destructive Control of Complex Rhythmic Systems." (June 2007)

3. U.S. provisional patent # 60/456,840, International provisional patent # PCT/US2004/008611, "Chemical Agent Alarm Monitor." (March 2004)

4. U.S. provisional patent # 60/316,813, "A Colloidal Pen for Producing Colloidal Crystals." (June 2001)

Participation in Business Development and Commercialization

Company Name: Medical Informatics Corp.
Location: Houston, TX
Funds Raised: ~$35,000,000 (Series A - Data Collective, Intel Capital. Series B – Intel Capital, Catalio Capital Management)
Full Time Employees: ~100
Description:

I co-founded Medical Informatics Corp. to provide a commercialization pathway for a technology called Sickbay, which enables real-time clinical monitoring and analytics in critical care environments. Sickbay was named the Forst and Sullivan Leader in virtual critical care in its most recent report, outperforming GE, Philips, and EPIC. The company has taken its main products through the FDA, receiving clearance as a class II 510(k) medical device. Several pieces of IP have been or are in the process of being licensed to Medical Informatics Corp from Baylor College of Medicine. Examples of licensed IP include an algorithm which predicts cardiorespiratory arrest in children with single ventricle physiology 1-2 hours before it happens. The company is currently in the process of taking this algorithm through the FDA. Currently, Medical Informatics has products installed in a significant number of hospitals across the country, including many high-profile institutions. Currently Sickbay watches over approximately 2% of all hospital beds in the USA. This provides Baylor College of Medicine with a large marketing opportunity to capitalize on the success of this venture.

III. TEACHING INFORMATION

A. Educational Leadership Roles (title, dates, responsibilities)

Student Teaching, Education & Mentorship

- Danielle Rios, MD (Assistant Professor)
  Department of Pediatrics - Neonatology, Baylor College of Medicine
  Roles: Research Mentor 2015 - present, NIH K Award Mentor, 2018 - present
  Responsibilities: As Dr. Rios's mentor, I provide oversight and experience to ensure that her research program is achieving its goals and objectives, and help her overcome barriers to progress that may be encountered. Dr. Rios's research area of focus is the understanding of deterioration of ELBW infants in the early days of life. I assisted in the development and scoping of her K Award which was funded by the NIH. As her K award mentor, I also provide accountability and oversight to ensure that Dr. Rios is meeting the milestones outlined in her application. This involved weekly meetings, technical and mathematical scoping, manuscript preparation, presentation preparation, as well as assistance in data collection and large-scale numerical data analysis.

- Chris Rhee, MD (Assistant Professor)
  Department of Pediatrics - Neonatology, Baylor College of Medicine
  Roles: Research Mentor 2014 - present, NIH K Award Mentor, 2018 - present
  Responsibilities: As Dr. Rhee's mentor, I provide oversight and experience to ensure that his research program is achieving its goals and objectives. Dr. Rhee's research focuses on improving blood pressure management of neonatal infants by measuring critical closing pressure in the cerebral vasculature. I help Dr. Rhee overcome barriers to data collection that are often encountered with such projects. I have assisted in the development and scoping of Dr. Rhee's K Award which was funded by the NIH. I ensure that the milestones and training objectives set out this award are

met. This involved weekly meetings, technical and mathematical scoping, manuscript preparation, presentation preparation, as well as assistance in data collection and large-scale numerical data analysis.

- Eric Vu, MD (Assistant Professor)
  Department of Anesthesiology, Baylor College of Medicine
  Role: Mentor, 2015 - present
  Responsibilities: As Dr. Vu's mentor, I provide guidance to ensure that his research program is achieving its goals and objectives. Dr. Vu's research focus is on predicting deterioration in patients with heart disease. I have helped Dr. Vu overcome barriers to data collection that are often encountered with data science projects. I helped build a rigorous course curriculum for Dr. Vu's masters degree in clinical data science, with emphasis on statistics, computer programming, machine learning and database architecture. I helped Dr. Vu in the development and scoping of several grant proposals, 2 of which were funded by extramural foundations. I ensure that the milestones and objectives set out these awards are met. This involved weekly meetings, technical and mathematical scoping, manuscript preparation, presentation preparation, as well as assistance in data collection and large-scale numerical data analysis.

- Sebastian Acosta, Ph.D. (Post-doctoral fellow)
  Role: Advisor, 2012 - present
  Responsibilities: I have been Dr. Acosta's advisor and mentor since he joined my research group as a post-doctoral fellow in 2012. I provide the leadership, oversight and experience to ensure that our research group's objectives are achieved. This involved weekly meetings, technical and mathematical scoping, manuscript preparation, presentation preparation, as well as assistance in data collection and large-scale numerical data analysis. Dr. Acosta's work has significantly improved in the area of medical research. Because of my mentorship to Dr. Acosta, he has been promoted to the rank of Instructor in the Department of Pediatrics at BCM. Multiple joint peer-reviewed publications have been produced, and a track record of extramural funding has been established. Dr. Acosta also has successfully competed for NSF funding launching his independent scientific career.

- Charles Puelz (Graduate student)
  Role: Doctoral Advisor, 2012 - 2017
  Degree: Ph.D. Computational and Applied Mathematics, Rice University
  Dissertation Title: Numerical Methods and Applications for Reduced Models of Blood Flow
  Responsibilities: I was Dr. Puelz's doctoral advisor and mentor since he joined my research group as a graduate student in 2012. Dr. Puelz's research focus area was on closed-loop modeling the hemodynamic circuit of a human in real-time. I provided the leadership, oversight and experience to ensure that his research goals were achieved and that he received his doctoral degree. This involved multiple weekly meetings, technical and mathematical project scoping, manuscript preparation, and academic presentation preparation. We have co-authored several peer-reviewed publications on modeling the cardiovascular system, including a new equation solver for real-time blood-flow simulations. Because of my mentorship to Dr. Puelz, he was awarded a prestigious 3-year post-doctoral fellowship at University of North Carolina Chapel Hill and New York University, which is the #1 Applied Math department in USA, by USNWR.

- Emily Hendryx (Graduate student)
  Role: Doctoral Advisor, 2012 - 2018
  Degree: Ph.D. Computational and Applied Mathematics, Rice University
  Dissertation Title: Subset Selection and Feature Identification in the Electrocardiogram
  Responsibilities: I was Dr. Hendryx's doctoral advisor and mentor since she joined my research group as a graduate student in 2012. Dr. Hendryx's research focus was on ECG morphology selection using unsupervised machine learning techniques. I provided the leadership, oversight and

experience to ensure that her research goals were achieved and that she received her doctoral degree. This involved multiple weekly meetings, technical and mathematical project scoping, manuscript preparation, presentation preparation, as well as assistance in data collection and large-scale numerical data analysis. We have co-authored several peer-reviewed publications on subset selection of representative ECG morphologies. Because of my mentorship, she received 4 tenure-track offers for university teaching positions, ultimately choosing a faculty position within the department of Mathematics at University of Central Oklahoma.

- Risa Myers (Graduate student)
  Role: Academic Mentor & Dissertation defense committee member, 2013- 2016
  Degree: Ph.D. Computer Science, Rice University
  Dissertation Title: Synthesis of Patient Data to Predict Outcomes
  Responsibilities: I was one of Dr. Myers' doctoral advisors and academic mentors when while she worked with my research group as a graduate student in 2013. Dr. Myers' research focus was on using physiologic data to predict bad outcomes in critically ill patients. I provided the leadership, oversight and experience to ensure that her research goals were achieved and that she received her doctoral degree. This involved multiple weekly meetings, technical and mathematical project scoping, manuscript preparation, presentation preparation, as well as assistance in data collection and large-scale numerical data analysis. We have co-authored a high-profile paper predicting crisis events in patients with TBI which was published in Critical Care Medicine. After graduating Dr. Myers accepted a teaching faculty in the department of Computer Science at Rice University.

- Anthony Nguyen (Undergraduate student - Summer Intern)
  Role: Mentor, 2012
  Degree: BS. Computational and Applied Mathematics, Rice University
  Responsibilities: I was Mr. Nguyen's summer research advisor when he joined my research group as an undergraduate student in 2012. Mr. Nguyen's research involved the mathematical analysis of ECG & EEG waveform recordings for quantifying cerebral-cardiac synchronization during administration of different types of medications.

- Jay Achuff (Undergraduate student - Summer Intern)
  Role: Summer Research Advisor & Mentor, 2015
  Degree: BS. Computer Science and Applied Mathematics, Carnegie Mellon University
  Responsibilities: I was Mr. Achuff's summer research advisor when he joined my research group as an undergraduate student in 2015. Mr. Achuff's work involved the development of a computer algorithm to analyze physiologic time series data to look for outlier patterns with a known morphology. After graduating with his BS degree, he joined Tableau as a full-time software developer.

- Hannah Park (Undergraduate student - Summer Intern & Independent research student)
  Role: Research Advisor, 2016
  Degree: BS. Computational and Applied Mathematics, Rice University
  Responsibilities: I was Ms. Park's summer and independent research advisor when she joined my research group as an undergraduate student in 2016. Ms. Park's work involved the development of a computer algorithm to analyze physiologic time series data after administrations of medications to look for patterns associated with adverse events. This work resulted in 2 manuscripts which found that common clinical medications have unexpected physiologic consequences. After graduation, Ms. Park began a graduate program in applied mathematics.

- Anna Norris (Undergraduate student - Summer Intern & Independent research student)
  Role: Summer & Independent Research Advisor, 2016
  Degree: BS. Mechanical Engineering, Rice University

Responsibilities: I was Ms. Norris' summer and independent research advisor when she worked with my research group as an undergraduate student in 2016. Ms. Norris' work involved the development of an acoustic sensor which could detect the "roar" within the lung of a neonatal infant when they their reparation is supported by a CPAP device. This could be used to ensure that the nasal interface was not obstructed, preventing potential failure onto mechanical ventilation. I was responsible for project scoping, guidance, and oversight for Ms. Norris while she developed the prototype device. This involved monthly meetings, technical and mathematical scoping, as well as assistance in data collection and numerical data analysis, and device testing and validation.

- Colin Edwards
  Department of Computational and Applied Mathematics, Rice University
  Roles: Research Mentor: 2021-2022
  I was Mr. Edwards research mentor for his summer internship and independent research coursework while he was finishing up his degree at Rice University. Colin's project involved creating 3D meshes of the heart and arch of children with congenital heart disease from MRI and CT imaging studies as well as making measurements of the geometric properties of study subjects.

- Kristina Montez
  Department of Chemical and Biological Engineering, Rice University
  Roles: Research Mentor: 2022-2023
  I was MS. Montez's research mentor for her summer internship and independent research coursework while she was finishing up her engineering degree at Rice University. Kristina's project involved creating 3D meshes of the heart and arch of children with congenital heart disease from MRI and CT imaging studies as well as making measurements of the geometric properties of study subjects. I provided direction and oversight to her to ensure a successful research experience. She was able to use this work to gain acceptance to the UT school of medicine in their prestigious EnMed program. This involved weekly meetings with her, technical and mathematical scoping, as well as assistance in data collection and analysis of the resulting data sets, as well as manuscript preparation.

- Dan Lior, PhD
  Department of Pediatrics - Cardiology, Baylor College of Medicine
  Roles: Research Mentor 2019 - present
  As Dr. Lior primary mentor, I provide oversight and expertise to ensure that his work is progressing according to the requirements of his research projects. Dr. Lior is focused on 3D meshing and structural modeling of cardiovascular system of children with different types of congenital heart disease. His work has resulted in several peer-reviewed publications and grant submissions. My mentorship involves weekly meetings, technical and mathematical scoping, as well as assistance in data collection and numerical data analysis.

- Minh Nyugen, MD (Assistant Professor)
  Department of Pediatrics - Cardiology, Baylor College of Medicine
  Roles: Research Mentor 2022 - present
  Responsibilities: As Dr. Nyugen's mentor, I provide oversight and experience to ensure that his research program is achieving its goals and objectives and help him overcome barriers to progress that may be encountered. Dr. Nyugen's research area of focus is the application of machine learning techniques to large-scale echo imaging data to better predict diastolic disfunction in children with hypertrophic cardiomyopathy. I assisted in the development and scoping of his K Award which is currently under review by the NIH. As his K award mentor, I also provide accountability and oversight to ensure that Dr. Nyugen is meeting the milestones outlined in his project timeline. This involved weekly meetings, technical and mathematical scoping, manuscript

preparation, presentation preparation, as well as assistance in data collection and large-scale numerical data analysis.

Small-Group Courses

- Rice Senior Design Course CAAM 495/496
  Students: Alice Xie, AJ Velasquez Mao, Jennifer Dawkins, Chris Sabbagh, Tony Ren (Undergraduate students - Capstone Senior Design Project), Rice University.
  Project: CPAP Roar
  Role: Faculty Advisor, 2016 - 2017
  Responsibilities: I was the project advisor for the CPAP roar project which was run out of the Oshman Engineering Design Kitchen at Rice University. This capstone team was made up of 5 senior undergraduate students from across Rice University. The project involved the continued development of an acoustic sensor which could detect the "roar" within the lung of a neonatal infant when they their reparation is supported by a CPAP device. This could be used to ensure that the nasal interface was not obstructed, preventing potential failure onto mechanical ventilation. I was responsible for scoping the project and keeping the team on track to complete the project objectives. This involved weekly meetings, technical and mathematical scoping, presentation preparation, as well as assistance in data collection and numerical data analysis.

- Rice Senior Design Course CAAM 495/496
  Students: Muhammad Shamim, Pakorn Wongwaitayakornkul
  (Undergraduate students - Capstone Project), Rice University
  Project: Inter-cardiac blood pressure estimation using ultrasound
  Role: Faculty Advisor, 2013-2014
  Responsibilities: I was the project advisor for the Inter-cardiac blood pressure monitoring capstone project. This capstone team was made up of 4 senior undergraduate students from the Computational and Applied Mathematics department of Rice University. The goal of the project was to estimate the pressure gradients inside the heart using blood flow velocity changes as measured using a 2D echocardiogram. I was responsible for scoping the project and keeping the team on track to complete the project objectives. This involved weekly meetings, technical and mathematical scoping, presentation preparation, as well as assistance in data collection and numerical data analysis. Group members went on to medical school at BCM and a PhD program in applied physics at CalTech.

- Rice Senior Design Course CAAM 495/496
  Students: Andrew Tilley, Joey Huchette, Hrothgar, Guang Yang
  (Undergraduate students - Capstone Project), Rice University
  Project: Inter-cardiac blood pressure estimation using ultrasound
  Role: Faculty Advisor, 2012-2013
  Responsibilities: I was the project advisor for the Inter-cardiac blood pressure monitoring capstone project. This capstone team was made up of 4 senior undergraduate students from the Computational and Applied Mathematics department of Rice University. The goal of the project was to estimate the pressure gradients inside the heart using blood flow velocity changes as measured using a 2D echocardiogram. I was responsible for scoping the project and keeping the team on track to complete the project objectives. This involved weekly meetings, technical and mathematical scoping, presentation preparation, as well as assistance in data collection and numerical data analysis. Members of the group went on to prestigious PhD programs at MIT, Berkley, and Oxford.

IV. PATIENT CARE AND CLINICAL CONTRIBUTIONS

A. Patient Care Responsibilities

I have no patient care responsibilities as I am not a physician.

B. Clinical Leadership or Business Development

I have no clinical leadership responsibilities as I am not a physician.

C. Voluntary Health Organization Participation

I have no voluntary health organization participation.

D. Contributions, Initiatives or Recognition Related to Patient Safety and Healthcare Quality (include information about implementation, outcomes or dissemination, where applicable)

Project: Sickbay Implementation across Texas Children's Hospital

Implementation: Sickbay went live at Texas Children's Hospital for clinical use in 2016. It is now in over 360 beds across the institution including ICU, step-down units, ORs and ED beds.

Outcome: Sickbay has over 1,500 user interactions per day (as of Oct 1, 2018) with approximately 600 registered user accounts. Multiple patient saves have been reported by physicians and nurses thanks for the high-resolution data that is collected by the system.

National Recognition: A dozen hospitals (adult and pediatric) across the US now have Sickbay installed, including some of the top institutions in the country.

Project: Alarm Management in the PCU at Texas Children's Hospital

Implementation: The Sickbay platform was used to identify patients with abnormal levels of alarms. The sickbay platform was then used to optimize the alarm settings for these patients according to a clinical protocol adopted by the PCU care team.

Outcome: 70% reduction in alarms in the unit. 50% reduction in nurse time in flood. For reference, nurse time in flood is a direct measure of the risk of a nurse missing an alarm.

National Recognition: 2016 AAMI national award. 2017 ECRI national award.

E. Contributions to Health and Science Policy

I do not have contributions to Health and Science Policy.

V. SERVICE CONTRIBUTIONS

Texas Children's Hospital Committees

Sickbay Advisory Committee - This committee was formed to provide oversight related to the implementation and rollout of the Sickbay analytics platform across Texas Children's Hospital.

Content and Analytics Committee - This committee was formed to advise Texas Children's Hospital regarding data science and data management for research and quality projects.

Research Operations Committee (Heart Center) - This committee was formed to advise Texas Children's Hospital heart center leadership regarding data science and data management for research and quality projects.

Baylor College of Medicine Committees

IP Sub-committee - This committee was formed to advise the in-Chief for Research at BCM/TCH with regards to improvements in the IP/Licensing/Commercialization efforts of BCM and TCH.

A. Administrative Assignments and Committees

I have no administrative assignments.

B. National, Regional or Local Participation in Professional or Voluntary Organizations.

None.

C. Other Pertinent Information

None.

March 30, 2024